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Hafþór Árni Hermannsson

Mechatronics Engineer — Prototyping, Robotics & Field Hardware

Hands-on mechatronics engineer who enjoys designing, building and experimenting both professionally and in my free time. Years of self-directed projects across electronics, embedded systems, mechanisms and fabrication, alongside two years developing camera and laser hardware for deployment on working fishing gear. Relocating to Amsterdam; available immediately.

Experience

Optitog May 2024 – present
Mechatronics Engineer · early-stage marine-tech startup – Reykjavík, Iceland

- Led the electronics and embedded hardware for underwater camera and laser modules deployed on working fishing trawls, including sensor and actuator integration, power regulation and thermal management.
- Main designer and maintainer of the custom 4-layer PCBs (EasyEDA), owning every revision after the initial implementation and integrating the power, sensing, control and Raspberry Pi interfaces for the deployed systems.
- Worked in a two-person team during deployments in Norway and Sweden, assembling and commissioning systems and troubleshooting, upgrading and replacing hardware in the field.
- Built Python pipelines pairing YOLO detections with sensor logs into the statistical reports that steered the next hardware revision; co-developed an event-camera laser unit under an Icelandic Student Innovation Fund grant.

Media ehf Summer 2023
Mechatronics Engineer – Reykjavík, Iceland

- Developed a ceiling-mounted mmWave radar prototype for non-contact elderly monitoring, integrating fall detection and vital-sign monitoring with microcontrollers, radar modules, custom electronics and 3D-printed housings.

Selected Projects

Internal Wave-Energy Harvester — M.Sc. Thesis publication 2026

- Designed, built and experimentally validated a bench-scale internal wave-energy harvester for a self-powered monitoring buoy, using a translating three-coil carriage on a segmented magnetic rod to convert buoy motion directly into electrical power.
- Built the complete experimental platform around it, including a custom coil winder, ESP32-S2 onboard motion/power instrumentation, and a rope-driven wave simulator with a 50 Hz PID + feedforward position loop, wheel odometry, operating-state machine, soft limits, watchdog and latched safety shutdown.
- Developed and validated coupled electromagnetic and mechanical models against measured experiments, then used them to explore scaled designs toward a 10 W buoy power target; the best design within the strict 1.8 m package envelope predicted about 8.8 W.

Invisible Drum Set — real-time embedded system GitHub 2024

- Dual-IMU motion-controlled drum kit on ATmega328P: bare-metal C drivers written from scratch (I2C, UART, ADC, timers), sensor fusion with drift recalibration, 30–50 ms motion-to-sound latency.

NATO TIDE Hackathon — casualty routing demo GitHub 2026

- Built a working casualty-evacuation routing demo within the hackathon window: hospital assignment by role of care and bed availability, with map-based route visualisation for operators.

Practical Work Experience

Summer and seasonal work since 2015, including **workshop assistant** at Vélfang, assembling and installing machinery; **deckhand** on a Norwegian longliner; **painter** at Nestak; **plumber's assistant** at Thor, including laying over 40 km of snow-melt piping beneath a football pitch; and three summers in fish processing at Síldarvinnslan.

Education

M.Sc. Mechatronics Engineering
Reykjavík University, 2023–2026
Graduated June 2026. Thesis: *Design and Optimisation of an Internal Wave Energy Harvester for a Self-Powered Monitoring Buoy*

B.Sc. Mechatronics Engineering
Reykjavík University, 2020–2023

Skills

Electronics: schematic capture and 4-layer PCB layout (EasyEDA, some KiCad), power regulation, motor and servo drive, sensor integration, soldering and rework, board bring-up and debugging

Mechanical & fabrication: Fusion 360 (some Inventor), 3D printing, waterjet cutting, workshop fabrication, mechanism design, component sourcing

Embedded & control: C/C++ (bare-metal and Arduino), ESP32, ATmega, STM32, nRF52840, Raspberry Pi and embedded Linux, I2C, SPI, UART, CAN; PID loops, state machines, safety interlocks and watchdogs

Software & simulation: Python (OpenCV, pandas), MATLAB/Simulink, Linux/Bash, YOLO; Ansys Maxwell, FEM and physics modelling

Self-Directed Builds

20+ documented home builds: Tesla coil (hand-wound secondary, DIY driver boards), 3D-printed star-tracking camera mount, induction heater, high voltage supply, IR theremin synthesizer

Autonomous RPi rover: UDP motor control, H.264 streaming, vision and LLM scene feedback · GitHub

nRF52840 2.4 GHz sniffer: reverse-engineering proprietary wireless HID protocols · GitHub

bestavedrid.is: live geospatial weather platform (Flask, Leaflet)

Practical

Icelandic — native
English — fluent
Danish — basic

Icelandic citizen (EEA) — no visa or sponsorship required in NL

Relocating to Amsterdam; onsite full-time; available immediately

Driving licence: B